

LAB EXPERIMENT – 2

8-Queen Problem using Python

Aim

To write a Python program to solve the **8-Queen problem** using the **Backtracking algorithm**.

Procedure

1. Create an **8 × 8 chessboard**.
2. Place one queen in each row.
3. Check whether the position is safe from previously placed queens.
4. A queen must not share the same **column** or **diagonal** with another queen.
5. If the position is safe, place the queen and move to the next row.
6. If no safe position is available, **backtrack** to the previous row.
7. Continue until all 8 queens are placed successfully.
8. Display the final chessboard arrangement.

Source Code

main.py

Visualize

Run ▶

```
1 N = 8
2
3 board = [[0 for _ in range(N)] for _ in range(N)]
4
5
6 def is_safe(row, col):
7     # Check column
8     for i in range(row):
9         if board[i][col] == 1:
10             return False
11
12     # Check upper-left diagonal
13     i = row - 1
14     j = col - 1
15
16     while i >= 0 and j >= 0:
17         if board[i][j] == 1:
18             return False
19         i -= 1
20         j -= 1
21
22     # Check upper-right diagonal
23     i = row - 1
24     j = col + 1
25
26     while i >= 0 and j < N:
27         if board[i][j] == 1:
28             return False
29         i -= 1
30         j += 1
31
32     return True
33
34
35 def solve(row):
36     if row == N:
37         return True
38
39     for col in range(N):
40
41         if is_safe(row, col):
42             board[row][col] = 1
43
44             if solve(row + 1):
45                 return True
46
47             # Backtracking
48             board[row][col] = 0
49
50     return False
51
52
53 if solve(0):
54     print("Solution for 8-Queen Problem:\n")
55
56     for row in board:
57         print(" ".join("Q" if cell == 1 else "." for cell in row))
58 else:
59     print("No solution exists.")
```

Output

Solution for 8-Queen Problem:

Q
. . . . Q . . .
. Q
. Q . .
. . Q
. Q .
. Q
. . . Q

Result

Thus, the 8-Queen problem was successfully solved using the Backtracking algorithm in Python.